

8.1.1 Solving linear systems by solving a minimization problem

8.1.1 Solving linear systems using minimization, Part 1

fit width



Consider the quadratic polynomial

$$f(\chi) = \frac{1}{2}\alpha\chi^2 - \beta\chi.$$

Finding the value $\hat{\chi}$ that minimizes this polynomial can be accomplished via the steps:

- Compute the derivative and set it to zero:

$$f'(\hat{\chi}) = \alpha\hat{\chi} - \beta = 0.$$

We notice that computing $\hat{\chi}$ is equivalent to solving the linear system (of one equation)

$$\alpha\hat{\chi} = \beta.$$

- It is a minimum if $\alpha > 0$ (the quadratic polynomial is concaved up).

Obviously, you can turn this around: in order to solve $\alpha\hat{\chi} = \beta$ where $\alpha > 0$, we can instead minimize the polynomial

$$f(\chi) = \frac{1}{2}\alpha\chi^2 - \beta\chi.$$

This course does not have multivariate calculus as a prerequisite, so we will walk you through the basic results we will employ. We will focus on finding a solution to $Ax = b$ where A is symmetric positive definite (SPD). (In our discussions we will just focus on real-valued problems). Now, if

$$f(x) = \frac{1}{2}x^T Ax - x^T b,$$

then its **gradient** equals

$$\nabla f(x) = Ax - b.$$

The function $f(x)$ is minimized (when A is SPD) when its gradient equals zero, which allows us to compute the vector for which the function achieves its minimum. The basic insight is that in order to solve $A\hat{x} = b$ we can instead find the vector $\hat{x}$ that minimizes the function $f(x) = \frac{1}{2}x^T Ax - x^T b$.

8.1.1 Solving linear systems using minimization, Part 2

fit width



Theorem 8.1.1.1. *Let A be SPD and assume that $A\hat{x} = b$. Then the vector $\hat{x}$ minimizes the function $f(x) = \frac{1}{2}x^T Ax - x^T b$.*

Proof.

This proof does not employ multivariate calculus!

Let $A\hat{x} = b$. Then

$$\begin{aligned}
f(x) &= \text{< definition of } f(x) \text{ >} \\
&\frac{1}{2}x^T Ax - x^T b \\
&= \text{< } A\hat{x} = b \text{ >} \\
&\frac{1}{2}x^T Ax - x^T A\hat{x} \\
&= \text{< algebra >} \\
&\frac{1}{2}x^T Ax - x^T A\hat{x} + \underbrace{\frac{1}{2}\hat{x}^T A\hat{x} - \frac{1}{2}\hat{x}^T A\hat{x}}_0 \\
&= \text{< factor out >} \\
&\frac{1}{2}(x - \hat{x})^T A(x - \hat{x}) - \frac{1}{2}\hat{x}^T A\hat{x}.
\end{aligned}$$

Since $\hat{x}^T A\hat{x}$ is independent of x , and A is SPD, this is clearly minimized when $x = \hat{x}$.